

## Extension implemented:

- Isomax loss
- Logits normalization
- LORA support
- Dataset on Cityscapes with outlier exposure from COCO
- RbA loss for Outlier Supervisioning

## Results:

### Baselines

#### ERFNET

|        |             |                      | SMIYC RA-21 |       | SMIYC RO-21 |       | FS L&F |       | FS Static |       | Road Anomaly |       |
|--------|-------------|----------------------|-------------|-------|-------------|-------|--------|-------|-----------|-------|--------------|-------|
| Model  | Method      | mIoU<br>(cityscapes) | AuPRC       | FPR95 | AuPRC       | FPR95 | AuPRC  | FPR95 | AuPRC     | FPR95 | AuPRC        | FPR95 |
| ERFNET | MSP         | 0.7217               | 29.08       | 62.57 | 2.71        | 65.20 | 1.73   | 50.61 | 5.03      | 40.91 | 12.42        | 82.59 |
|        | MaxLogit    |                      | 38.29       | 59.38 | 4.63        | 48.44 | 3.28   | 45.49 | 6.27      | 39.66 | 15.58        | 73.28 |
|        | Max Entropy |                      | 30.95       | 62.68 | 3.05        | 65.89 | 2.56   | 50.19 | 5.89      | 40.64 | 12.67        | 82.76 |

The metrics for ERFNet and EoMT are not directly comparable because the two models produce structurally different outputs (pixel logits vs. mask logits). As a consequence, the computation pipelines for AuPRC and FPR95 naturally differ. We kept the original evaluation criteria provided in the repository.

(all fine tuning was done with AMP enabled and masked attention disabled)

#### eomt\_base\_640 (prof weights):

|       |             |                      | SMIYC RA-21  |              | SMIYC RO-21  |             | FS L&F      |              | FS Static    |              | Road Anomaly |              |
|-------|-------------|----------------------|--------------|--------------|--------------|-------------|-------------|--------------|--------------|--------------|--------------|--------------|
| Model | Method      | mIoU<br>(cityscapes) | AuPRC        | FPR95        | AuPRC        | FPR95       | AuPRC       | FPR95        | AuPRC        | FPR95        | AuPRC        | FPR95        |
| EoMT  | MSP         | 0.817                | <b>67.91</b> | 13.46        | <b>85.44</b> | <b>3.94</b> | 16.36       | 12.96        | 58.17        | <b>41.15</b> | 67.80        | 23.89        |
|       | MaxLogit    |                      | 67.72        | <b>13.26</b> | 85.36        | 4.06        | 16.34       | <b>12.73</b> | 58.27        | 46.35        | 67.25        | 23.59        |
|       | Max Entropy |                      | 69.03        | 13.02        | 85.32        | 4.05        | <b>19.0</b> | 12.79        | 56.81        | 41.75        | <b>70.89</b> | 23.38        |
|       | RbA         |                      | 67.54        | 95.13        | 85.01        | 99.8        | 16.48       | 9.094        | <b>60.56</b> | 77.47        | 66.42        | <b>21.21</b> |

eomt\_large\_1024

(hugging face weights):

|       |             |                      | SMIYC RA-21  |              | SMIYC RO-21  |              | FS L&F       |              | FS Static    |              | Road Anomaly |              |
|-------|-------------|----------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| Model | Method      | mIoU<br>(cityscapes) | AuPRC        | FPR95        | AuPRC        | FPR95        | AuPRC        | FPR95        | AuPRC        | FPR95        | AuPRC        | FPR95        |
| EoMT  | MSP         | 0.8417               | 92.23        | 2.334        | 97.12        | 0.037        | 24.87        | 10.46        | 71.76        | 25.28        | 80.15        | 11.85        |
|       | MaxLogit    |                      | <b>92.53</b> | <b>2.329</b> | 97.12        | 0.036        | 25.05        | 10.24        | 71.88        | 24.12        | 80.51        | 11.77        |
|       | Max Entropy |                      | 88.24        | 2.328        | 97.03        | 0.034        | 24.71        | 10.29        | 68.63        | 24.93        | 73.01        | 11.64        |
|       | RbA         |                      | 92.42        | 2.330        | <b>97.18</b> | <b>0.032</b> | <b>25.31</b> | <b>6.023</b> | <b>73.54</b> | <b>14.48</b> | <b>80.82</b> | <b>11.29</b> |

### Temperature scaling

|                  |                      | SMIYC RA-21 |       | SMIYC RO-21 |       | FS L&F |       | FS Static |       | Road Anomaly |       |
|------------------|----------------------|-------------|-------|-------------|-------|--------|-------|-----------|-------|--------------|-------|
| Method           | mIoU<br>(cityscapes) | AuPRC       | FPR95 | AuPRC       | FPR95 | AuPRC  | FPR95 | AuPRC     | FPR95 | AuPRC        | FPR95 |
| MSP              | 0.7217               | 29.08       | 62.57 | 2.71        | 65.20 | 1.73   | 50.61 | 5.03      | 40.91 | 12.42        | 82.59 |
| MSP(t = 0.5)     |                      | 27.05       | 62.79 | 2.42        | 66.18 | 1.27   | 46.66 | 4.48      | 41.25 | 12.19        | 81.26 |
| MSP(t = 0.75)    |                      | 28.14       | 62.51 | 2.57        | 64.06 | 1.48   | 51.74 | 4.73      | 41.51 | 12.32        | 82.34 |
| MSP(t = 1.1)     |                      | 29.39       | 62.67 | 2.77        | 65.82 | 1.84   | 50.21 | 5.17      | 40.71 | 12.46        | 82.75 |
| MSP (best t) = 2 |                      | 30.59       | 65.06 | 3.01        | 72.62 | 3.03   | 46.40 | 6.99      | 41.77 | 12.66        | 86.43 |

MSP with T=1.0 is the baseline (without temperature scaling). The best T varies by dataset: T=2.0 for SMIYC datasets, and T=3.0 for Fishyscapes and RoadAnomaly. Temperature scaling improves AuPRC across all datasets, with gains of 1-2% on average, at the cost of slightly increased FPR95.

## OOD exposure:

eomt\_large\_1024

(hugging face weights + 8 epoch fine tuning class\_head and mask\_head with OOD supervision ( $p=0.5$ ,  $a = 5$  )):

|       |        |                   | SMIYC RA-21 |       | SMIYC RO-21 |        | FS L&F |       | FS Static |       | Road Anomaly |       |
|-------|--------|-------------------|-------------|-------|-------------|--------|--------|-------|-----------|-------|--------------|-------|
| Model | Method | mIoU (cityscapes) | AuPRC       | FPR95 | AuPRC       | FPR95  | AuPRC  | FPR95 | AuPRC     | FPR95 | AuPRC        | FPR95 |
| EoMT  | RbA    | 0.832             | 90.77       | 2.452 | 93.01       | 0.7538 | 26.79  | 6.125 | 76.90     | 7.549 | 80.40        | 11.14 |

eomt\_large\_1024

(hugging face weights + 8 epoch fine tuning class\_head and mask\_head + Lora with OOD supervision ( $p=0.5$ ,  $a = 5$ ,  $k = 10^{-5}$  )):

|       |        |                   | SMIYC RA-21 |       | SMIYC RO-21 |       | FS L&F |       | FS Static |       | Road Anomaly |       |
|-------|--------|-------------------|-------------|-------|-------------|-------|--------|-------|-----------|-------|--------------|-------|
| Model | Method | mIoU (cityscapes) | AuPRC       | FPR95 | AuPRC       | FPR95 | AuPRC  | FPR95 | AuPRC     | FPR95 | AuPRC        | FPR95 |
| EoMT  | RbA    | 0.8413            | 92.35       | 2.367 | 95.95       | 0.743 | 25.57  | 4.771 | 89.39     | 1.832 | 82.86        | 9.385 |

eomt\_large\_1024

(hugging face weights + 8 epoch fine tuning class\_head and mask\_head + Lora with OOD supervision ( $p=1$ ,  $a = 5$ ,  $k = 10^{-3}$  )):

|       |        |                   | SMIYC RA-21 |       | SMIYC RO-21 |       | FS L&F |       | FS Static |       | Road Anomaly |       |
|-------|--------|-------------------|-------------|-------|-------------|-------|--------|-------|-----------|-------|--------------|-------|
| Model | Method | mIoU (cityscapes) | AuPRC       | FPR95 | AuPRC       | FPR95 | AuPRC  | FPR95 | AuPRC     | FPR95 | AuPRC        | FPR95 |
| EoMT  | RbA    | 0.8414            | 92.58       | 2.351 | 95.60       | 0.750 | 27.49  | 5.14  | 93.97     | 1.086 | 85.40        | 9.565 |

eomt\_base\_640

(hugging face weights + 8 epoch fine tuning class\_head and mask\_head + Lora with OOD supervision ( $p=1$ ,  $a = 5$ ,  $k = 10^{-3}$  )):

|       |        |                   | SMIYC RA-21 |       | SMIYC RO-21 |       | FS L&F |       | FS Static |       | Road Anomaly |       |
|-------|--------|-------------------|-------------|-------|-------------|-------|--------|-------|-----------|-------|--------------|-------|
| Model | Method | mIoU (cityscapes) | AuPRC       | FPR95 | AuPRC       | FPR95 | AuPRC  | FPR95 | AuPRC     | FPR95 | AuPRC        | FPR95 |
| EoMT  | RbA    | 0.817             | 65.24       | 87.79 | 92.44       | 1.57  | 19.55  | 7.133 | 89.15     | 3.171 | 73.77        | 11.72 |

Incorporating OOD objects from COCO into the Cityscapes dataset improves performance, particularly on the FS Static benchmark. This is due to the fact that FS Static is also

synthetically generated by overlaying images onto street scenes. While fine-tuning only the class and mask heads leads to a slight performance degradation on other dataset and semantic classification tasks, training also the final encoder layers via LoRA yields superior results compared to the baseline, while preserving the performance in semantic classification tasks.

## Isomax:

eomt\_large\_1024 with isomax

(hugging face weights + training of class\_head for 8 epochs):

|       |          |                      | SMIYC RA-21 |       | SMIYC RO-21 |        | FS L&F |       | FS Static |       | Road Anomaly |       |
|-------|----------|----------------------|-------------|-------|-------------|--------|--------|-------|-----------|-------|--------------|-------|
| Model | Method   | mIoU<br>(cityscapes) | AuPRC       | FPR95 | AuPRC       | FPR95  | AuPRC  | FPR95 | AuPRC     | FPR95 | AuPRC        | FPR95 |
| Eomt  | MaxLogit | 0.840                | 91.18       | 2.763 | 94.96       | 0.7861 | 24.24  | 30.22 | 70.54     | 68.10 | 81.10        | 9.498 |
|       | RbA      |                      | 91.02       | 2.825 | 95.64       | 0.1615 | 24.32  | 15.62 | 73.42     | 21.39 | 81.49        | 9.001 |

eomt\_base\_640 with logits norm

(prof weights + training of class\_head for 8 epochs):

|       |          |                      | SMIYC RA-21 |       | SMIYC RO-21 |       | FS L&F |       | FS Static |       | Road Anomaly |       |
|-------|----------|----------------------|-------------|-------|-------------|-------|--------|-------|-----------|-------|--------------|-------|
| Model | Method   | mIoU<br>(cityscapes) | AuPRC       | FPR95 | AuPRC       | FPR95 | AuPRC  | FPR95 | AuPRC     | FPR95 | AuPRC        | FPR95 |
| Eomt  | MaxLogit |                      | 67.80       | 75.44 | 59.93       | 99.99 | 16.90  | 56.69 | 54.90     | 77.65 | 61.57        | 48.45 |
|       | RbA      |                      | 53.03       | 98.87 | 53.65       | 99.99 | 17.22  | 60.01 | 54.41     | 99.30 | 59.90        | 91.90 |

Isomax does not improve performance.

## Logits norm:

eomt\_large\_1024 with logits norm

(hugging face weights + training of class\_head for 8 epochs):

|       |          |                      | SMIYC RA-21 |       | SMIYC RO-21 |       | FS L&F |       | FS Static |       | Road Anomaly |       |
|-------|----------|----------------------|-------------|-------|-------------|-------|--------|-------|-----------|-------|--------------|-------|
| Model | Method   | mIoU<br>(cityscapes) | AuPRC       | FPR95 | AuPRC       | FPR95 | AuPRC  | FPR95 | AuPRC     | FPR95 | AuPRC        | FPR95 |
| Eomt  | MaxLogit | 0.841                | 92.46       | 2.281 | 97.91       | 0.022 | 25.04  | 8.771 | 71.14     | 68.45 | 80.98        | 10.70 |
|       | RbA      |                      | 91.34       | 2.429 | 98.65       | 0.026 | 25.05  | 82.04 | 74.18     | 15.87 | 81.08        | 9.428 |

eomt\_base\_640 with logits norm  
(prof weights + training of class\_head for 8 epochs):

|       |          |                      | SMIYC RA-21 |       | SMIYC RO-21 |       | FS L&F |       | FS Static |       | Road Anomaly |       |
|-------|----------|----------------------|-------------|-------|-------------|-------|--------|-------|-----------|-------|--------------|-------|
| Model | Method   | mIoU<br>(cityscapes) | AuPRC       | FPR95 | AuPRC       | FPR95 | AuPRC  | FPR95 | AuPRC     | FPR95 | AuPRC        | FPR95 |
| Eomt  | MaxLogit |                      | 80.46       | 3.98  | 92.99       | 0.269 | 14.95  | 45.69 | 56.70     | 75.18 | 68.83        | 14.84 |
|       | RbA      |                      | 56.10       | 84.17 | 83.25       | 85.45 | 15.99  | 69.00 | 56.64     | 79.26 | 65.50        | 62.19 |

eomt\_base\_640 with logits norm  
(prof weights + 8 epoch fine tuning class\_head and mask\_head + Lora):

|       |          |                      | SMIYC RA-21 |       | SMIYC RO-21 |       | FS L&F |       | FS Static |       | Road Anomaly |       |
|-------|----------|----------------------|-------------|-------|-------------|-------|--------|-------|-----------|-------|--------------|-------|
| Model | Method   | mIoU<br>(cityscapes) | AuPRC       | FPR95 | AuPRC       | FPR95 | AuPRC  | FPR95 | AuPRC     | FPR95 | AuPRC        | FPR95 |
| Eomt  | MaxLogit |                      | 68.99       | 22.37 | 88.43       | 1.39  | 11.60  | 13.85 | 61.98     | 36.46 | 63.67        | 19.52 |
|       | RbA      |                      | 58.84       | 97.97 | 86.79       | 99.75 | 11.81  | 26.35 | 63.14     | 60.40 | 62.58        | 55.29 |

Logit normalization leads to an improvement in AuPRC, but an anomalous increase in FPR95 is observed for FS L&F with RBA and fs\_static with ML

Finetuning of class\_head, mask\_head and transformer block decrease performance respect only finetuning class\_head, with the exception of fs\_static